

Substrate SM Program — honest-state ledger v0.3 (Saturday afternoon batch absorption)

Keeper

2026-05-30 Saturday — late afternoon

Honest-state ledger v0.3 — Saturday afternoon batch absorption

Standing rule (Cal #34 STANDING, narrow framing): every cell marked DERIVED below is DERIVED modulo the one keystone bet (canonical-basis element of $U_q(B_2)$ IS a physical particle with that K-type). The headline-cap does not drop the conditionality.

v0.3 delta from v0.2

Seven tier-gates filed Saturday afternoon; one calibration ratified. No layer-tier promotions; one gate refinement at (b); engine v0.3 candidates pre-reserved; methodology stack +1.

Bulk-color gate (b) — gauge-phenomenology layer refinement

	v0.2 status	v0.3 status
(b) SU(3) QCD emergence	OPEN, multi-week; two routes (C) hidden Hardy-dynamics, (D) pure counting	OPEN, multi-week; route COMMITTED to (C+D) UNIFIED per Lyra bulk-color v0.4. (D) provides 3-fold structural count via SO(3) sub-vector; (C) provides non-abelian gauge emergence via Toeplitz algebra on $H^2(S)$. The 8 gluons would emerge as $8 = \dim \text{su}(3)$ Toeplitz operators acting on the 3 SO(3) labels.
Conditions on (b)	—	C1: Toeplitz-source citation (Upmeier/Berezin/Vasilevski) must be VERIFIED-CITED with theorem/page before any external use (Cal #33 STANDING). C2: AdS/CFT analogy demoted to “evocative parallel, not structural argument” in v0.5+.

Gate (b) remains OPEN multi-week; the route is now COMMITTED rather than two-option.

Engine layer — v0.3 §6/§7 candidates pre-reserved

Claim	v0.3 status
Engine v0.3 §6 candidate (Drinfeld double + CPT, Toy 3617)	STRUCTURAL pre-reserved; K1·2 audit anchor reserved; pending Elie engine v0.3 consolidation doc
Engine v0.3 §7 candidate (bulk-color algebraic, Toy 3620: $SO(5) \supset SO(3) \times SO(2)$, $5=N_c+rank$)	STRUCTURAL pre-reserved; K1·4 audit anchor reserved; cross-confirms bulk-color (D) $SO(3)$ sub-vector

Engine v0.2 PASS holds; v0.3 substantively addresses K1-K3 conditions via Toys 3609 + 3610 + 3617 + 3620.

Falsifier suite — three sharp quantitative tests in flight

Falsifier	Status	Cross-check
F1 PMNS substrate fractions (Elie Toy 3618)	3/3 within 1σ ; $rank \cdot N_c \cdot g = 42$, $N_c \cdot n_C^2 = 75$, $N_c = 3$ numerators; $N_{max} = 137$ denominator	LIVE — JUNO/DUNE 2025-2030 tighten to 2σ
T1 lepton-count $24 = 4 \times 2 \times 3$ (Lyra T1 v0.1)	Pure-QG layer ($\dim V_{(1/2,1/2)} = 4$ Dirac) MATCHED; dictionary-combined 24-count MATCHED	LIVE — triggers on 4th gen, $0\nu\beta\beta \rightarrow$ Majorana, non-doublet, SUSY partners, extra-K-type leptons
Bell sub-Tsirelson $1/2^{N_c} = 1/8$ (T2469 SCMP via Toy 3626)	Substrate prediction $S^2 \leq Tsirelson^2 - 1/8$	LIVE — falsifiable at current Bell precision $\sim 1\%$ on S (SP-30 Vienna response pending)

Three independent quantitative falsifiers of the keystone bet; all LIVE; all MATCHED currently.

Cross-anchors and over-determination findings

Finding	Tier	Source
6-domain $225 = (N_c \cdot n_C)^2$ fingerprint (Bergman + Ag Debye + Cu/Ag + Ag/Pb + cosmological c_{reg} + CKM)	V_{cb}	)
8/39 BOTH cluster in dim 8-18 coincides with SM-gauge dimensional content	STRUCTURAL observation	Grace Two-Route Scan v0.1; conditional on N1 null-model before Casey-named-principle elevation
$12 = 4$ -route over-determination ($n_C + g$; $rank + N_c + g$; $rank \cdot C_2$; $N_c \cdot rank^2$)	STRUCTURAL — strongest substrate-arithmetic finding	Two-Route Scan v0.1
18-cell substrate spine in 66-K-type Phase B table	STRUCTURAL backbone	Elie Toy 3614 $\rightarrow$ Grace Periodic Table v0.5+

L3 dynamics — Quasi-Eigentone Framework added

Claim	v0.3 status
Decay mechanism = Green coproduct on Hall algebra (engine v0.1/v0.2 base)	FRAMEWORK — β -decay (E3) generalizes to all SM decays; grading conservation gives Q,B,L automatically; structural distinction true / quasi / eigentone-in-vacuum
SM stable particles (e, ν , γ , p, gluon) = sector bottoms (true eigentones / eigentone-in-vacuum)	FRAMEWORK
SM unstable particles (μ , τ , n, W, Z, H, hadrons) = excited states with Green-coproduct decay	FRAMEWORK
Quantitative decay rates (kernel matrix elements, branching ratios, CP violation)	OPEN, multi-week

Pedagogical layer — Movie V0 and Vol 16 spine

Item	Status
Spectral Score Theoretical Layer v0.1 (Movie V0 backbone)	FRAMEWORK / ORGANIZATIONAL; ready for Grace/Elie artifact build; N1+N2 caveats (gluon + Higgs placements provisional)
Vol 16 Ch1 §3 v0.1 ($A_{\text{sub}} \leftrightarrow U_q(B_2)$ at operator level)	PEDAGOGICAL SYNTHESIS; absorbs genus-convention update (n_C=5 GENUS; C_2=6 Casimir; g=7 embedding); N1+N2 sharpening (color 8-vs-3 gap; 15-vs-26 generator count)

Methodology stack — Calibration #34 STANDING under narrow framing

Layer	Discipline
Cal #27	Get the tier right (peak-convergence brake)
Cal #29	Ask the right question shape
Cal #32	Get the parameter role right (value+role+source)
Calibration #33	Distinguish RECALLED from VERIFIED-CITED (Source-Verification, STANDING)
Calibration #34 (NEW, narrow framing)	Conditional-tag-with-headline discipline (STANDING; Methodology Index Q16)

Methodology stack now 19 layers (18 → 19 with Calibration #34 STANDING).

Open gates carrying forward

Gate	Status carrying forward
Bulk-color gate (b) — SU(3) gauge-phenomenology emergence	OPEN multi-week; route (C+D) COMMITTED; Toeplitz-source verification + AdS/CFT demotion are v0.5+ conditions
Bulk-color gate (a) — counting layer	ADDRESSED via SO(3) sub-vector; cross-confirmed by Engine v0.3 §7 candidate ($5 = N_c + \text{rank}$)

Gate	Status carrying forward
Generation-mechanism identification	TENSION RELIEVED, NOT CLOSED; count-NUMBER forced ($h^v=N_c=3$); two-structures via bulk-color v0.2 (color geometric) + E7 spinor ³ (generations algebraic)
Per-particle quark/gauge dictionary (L2/L6) L4 v0.2 explicit per-particle mass derivations	ASSIGNED; flips when bulk-color closes per #418 UNBUILT; gated on Elie bulk K-type radial tower + Lyra L4 v0.2
L5 absolute scale + Higgs/EWSB Engine v0.3 consolidation doc	UNBUILT (least-developed L-axis) PENDING Elie filing (audit-support v0.2 was preparatory; the engine v0.3 doc itself with §6+§7 is the audit target)
A1 PRIMARY paper Cal quick-verify + proof-stage genus citation	PENDING — Keeper final PASS v0.4 already in place
C12-C14 T-number/META audit	PENDING (joint Lyra+Keeper)
Casey-named principle candidates from Two-Route Scan	BLOCKED on N1 null-model toy
SP-30 outreach drafts (eigentone, Cs-137, W-32)	PENDING Casey send-signals (#330)

Saturday batch summary

- 7 tier-gates filed Saturday afternoon: bulk-color v0.4, engine support v0.2, T1 falsifier v0.1, Two-Route Scan v0.1, Quasi-Eigentone v0.1, Spectral Score v0.1, Vol 16 §3 v0.1.
- 1 calibration ratified: Calibration #34 STANDING under narrow framing (headline-cap-conditionality).
- Saturday morning batch absorbed: 15 Elie toys (3612-3626); 6-domain 225 fingerprint; PMNS 3/3 within 1σ ; T1 falsifier; bulk-color v0.2/v0.3/v0.4 sequence; engine v0.3 §6/§7 candidates.
- Standing rule reaffirmed: every “DERIVED” carries modulo-keystone-bet conditional explicitly.
- Team state at EOD: standing reactive after substantial deep-work morning + structural Saturday afternoon; sustainable cadence held across all four CIs; queue substantively cleared.
- Multi-week work continues: bulk-color SU(3) emergence is the #1 macro frontier; engine v0.3 V-compound + L4 v0.2 + L5 absolute scale + B-track mathematical excavation pull in parallel per dateless work plan.

— Keeper. Honest-state ledger v0.3 absorbs Saturday afternoon batch + Calibration #34 STANDING; no layer-tier promotions; one gate refinement at (b); methodology stack 19 layers; queue ready for continuation work.